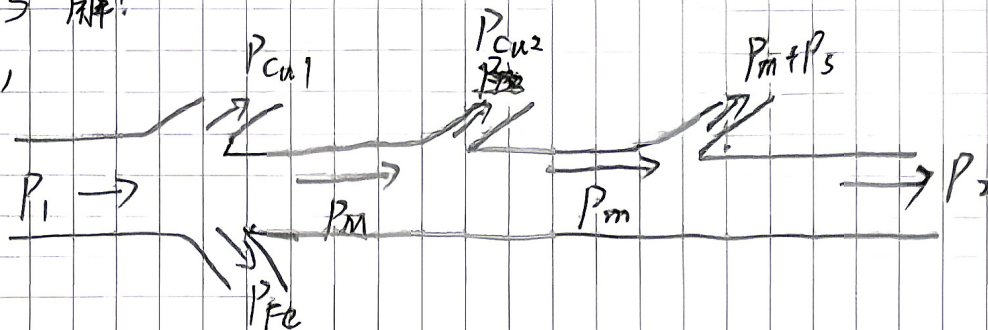


7.3 解:

1)



$$(2) P_N = P_2 = P_1 - P_{cu1} - P_{Fe} - P_{cu2} - P_m - P_s$$

$$\eta = \frac{P_N}{P_1} = \frac{5.5}{6.32} \times 100\% = 87\%$$

$$\begin{cases} P_{cu2} = s P_m \\ P_{cu2} = 237.5 W \end{cases}$$

$$s = \frac{P_{cu2}}{P_m} = 0.04$$

$$P_m = P_1 - P_{cu1} - P_{Fe} = 581.5 W$$

$$\text{又 } s = \frac{n_1 - n}{n_1}$$

假设该电机主磁极对数 $2p = 4$ 即 $p = 2$

$$\text{则 } n_1 = \frac{60f}{p} = 1500 \text{ r/min}$$

$$n = n_1 - s n_1 = 1440 \text{ r/min}$$

$$\text{电磁转矩 } T_1 = 9550 \frac{P_m}{n_1} = 955 \frac{P_m}{n_1} = 37 \text{ N}\cdot\text{m}$$

$$\text{输出转矩 } T_2 = 9550 \frac{P_N}{n} = 36.5 \text{ N}\cdot\text{m}$$

7.9 解:

$$(1) 2p = 8, p = 4$$

$$n_1 = \frac{60f}{p} = 750 \text{ r/min}$$

$$s = \frac{n_1 - n}{n_1} = 0.037$$

$$(2) T_N = 9550 \frac{P_N}{n_N} = 3439 \text{ N}\cdot\text{m}$$

$$(3) T_m = \lambda T_N = 7325.07 \text{ N}\cdot\text{m}$$

$$(4) s_m = \frac{s}{\lambda} (\lambda + \sqrt{\lambda^2 - 1}) = 0.148$$

$$\frac{T_1}{T_m} = \frac{\frac{2}{s_1}}{\frac{2}{s_m} + \frac{s_m}{s_1}}$$

$$(5) \frac{T_N}{T_1} = \frac{\frac{2}{s}}{\frac{2}{s_1} + \frac{s_1}{s}}$$

其中 $s_1 = 0.02$
为 $s_1 = 0.02$ 时的电磁转矩

$$= 0.837$$

$$T_1 = T_m \frac{\frac{2}{s_1}}{\frac{2}{s_m} + \frac{s_m}{s_1}}$$

$$= 1944 \text{ N}\cdot\text{m}$$

7.11 解:

$$1) \quad S = 0.025$$

$$S_m = S(\lambda + \sqrt{\lambda^2 - 1}) = 0.093$$

$$2) \quad 2p = 8 \quad p = 4$$

$$n_1 = \frac{60f_n}{p} = 750 \text{ r/min}$$

$$S = \frac{n_1 - n}{n_1}$$

$$n = n_1 - Sn_1 = 731.25 \text{ r/min}$$